Precept 7 Outline

ECO 202 TAs

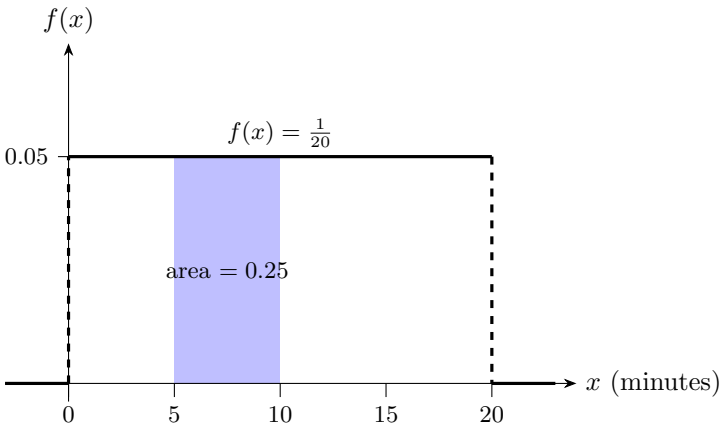
November 5, 2026

1 Weekly Check In

1. Any questions about the class, comments?
2. Exam 3 coming up this Monday

2 Precept practice problems

1. You walk up to a bus stop knowing a bus will arrive within the next 20 minutes, and every moment in that window is equally likely. Let X be your waiting time in minutes, so $X \sim \text{Uniform}(0, 20)$.



   (a) Write down the density $f(x)$ for all real x. Then check that the total area under f equals 1.

   (b) The density at $x = 5$ is $f(5) = 0.05$. Is this the probability that the bus arrives at exactly 5 minutes? What does the height $f(x)$ actually measure (think about its units)?

   (c) Find $\mathbb{P}(5 \le X \le 10)$ by computing the shaded area in the figure.

   (d) For $0 < h \le 15$, compute $\mathbb{P}(5 \le X \le 5 + h)$. Let $h \to 0$ to conclude what $\mathbb{P}(X = 5)$ is. Then explain why $\mathbb{P}(X < 5) = \mathbb{P}(X \le 5)$. Does "probability 0" mean "impossible"?

   (e) Let $Y \sim \text{Uniform}(0, \frac{1}{2})$. Find its density height and $\mathbb{P}(0.1 \le Y \le 0.2)$. How can a density be larger than 1 when probabilities cannot?

2. **(Practice 10, Question 5)** Two weekly cost shocks A and B, measured in hundreds of dollars, satisfy

$$\text{Var}(A) = 16, \qquad \text{Var}(B) = 9, \qquad \text{Cov}(A, B) = -6.$$

(a) Calculate $\mathrm{SD}(A)$, $\mathrm{SD}(B)$, and $\mathrm{Corr}(A, B)$.

(b) Calculate $\mathrm{Var}(A+B)$. Compare it with the answer obtained by incorrectly omitting covariance.

(c) If A and B were independent with the same marginal variances, what would $\mathrm{Cov}(A, B)$ and $\mathrm{Var}(A+B)$ be? Which implication are you using?

Now let U take values $-2, -1, 1, 2$ with probability $1/4$ each and let $V = \mathbf{1}\{|U| = 2\}$.

(d) Calculate $\mathbb{E}[U]$, $\mathbb{E}[V]$, $\mathbb{E}[UV]$, and $\mathrm{Cov}(U, V)$.

(e) Show directly that U and V are dependent despite their zero covariance.

(f) Calculate $\mathrm{Var}(U+V)$ with the variance-of-a-sum formula and verify it from the distribution of $U+V$.

3. **(Practice 13, Question 4)** A finite population has serial numbers $1, 2, 3, 4, 5$, so the unknown-target value in this fully visible teaching case is $m = 5$. Select an SRS of $n = 2$ distinct serial numbers without replacement. There are $\binom{5}{2} = 10$ equally likely samples.

   Consider the mean-based rule $T_{\text{mean}} = 2\bar{X} - 1$, the raw maximum M, and the corrected-maximum rule $T_{\max} = \frac{3}{2}M - 1$.

   (a) List all ten possible samples.

   (b) For each sample, calculate T_{mean}, M, and $T_{\max}$.

   (c) Construct the sampling distribution of each rule.

   (d) Calculate the expectation and bias of all three rules for $m = 5$.

   (e) Calculate the variance and MSE of the two unbiased rules.

   (f) Which unbiased rule has smaller MSE in this model? Explain why this finite example does not establish the same ranking under every alternative sampling mechanism or loss.

   (g) Explain why changing to independent sampling with replacement requires a new calculation rather than the same corrected-maximum formula.

4. A bag holds three tickets labeled 2, 4, and 6. A ticket is chosen at random, with each ticket equally likely. Let X be the number on one randomly chosen ticket, with population mean $\mu = \mathbb{E}[X]$ and population variance $\sigma^2 = \mathrm{Var}(X)$.

   We will study two statistics computed from a sample of two tickets, X_1 and X_2:

$$\overline{X} = \frac{X_1 + X_2}{2} \qquad \text{and} \qquad M = \max(X_1, X_2)$$

   The goal is to see when the usual formula $\mathrm{Var}(\overline{X}) = \sigma^2/n$ gives the right answer, and when it does not.

(a) Calculate μ and σ^2 for a single draw.

(b) Draw a ticket, record it, put it back, and draw again, so that X_1 and X_2 are independent. List all nine equally likely ordered samples (X_1, X_2) and, for each, calculate $\overline{X}$ and M.

(c) Construct the sampling distribution of $\overline{X}$ and of M, and verify that each set of probabilities sums to one.

(d) Using the sampling distribution of $\overline{X}$, calculate $\mathbb{E}[\overline{X}]$ and $\mathrm{Var}(\overline{X})$.

(e) Using the sampling distribution of M, calculate $\mathbb{E}[M]$ and $\mathrm{Var}(M)$.

(f) Calculate σ^2/n with $n = 2$. Compare it with $\mathrm{Var}(\overline{X})$ and with $\mathrm{Var}(M)$. For which statistic does the formula give the right answer?

(g) Now draw only *one* ticket X and record its value twice, giving the "sample" (X, X). Let $\overline{X} = (X + X)/2$. Find $\mathrm{Var}(\overline{X})$. Does it equal $\sigma^2/2$? Explain.

(h) Draw two tickets *without* replacement. List the possible samples (ignoring order), find the sampling distribution of $\overline{X}$, and calculate $\mathbb{E}[\overline{X}]$ and $\mathrm{Var}(\overline{X})$. Compare with $\sigma^2/2$.

(i) Fill in the table: for each scenario, does $\mathrm{Var} = \sigma^2/n = 4/3$ hold? Then state the two conditions that must both be satisfied before using σ^2/n.

Scenario	Statistic	Variance	Equals 4/3?
2 draws, with replacement	$\overline{X}$		
2 draws, with replacement	M		
1 draw, copied twice	$\overline{X}$		
2 draws, without replacement	$\overline{X}$		

3 Extra practice problems

1. Let Y take the values 0, 2, 4 with probabilities 0.30, 0.45, 0.25, and let $Y_1, \ldots, Y_{50}$ be independent random variables with the same distribution as Y.

   (a) Calculate $\mathbb{E}[Y]$ and $\mathrm{Var}(Y)$.

   (b) Use the central limit theorem, without a continuity correction, to approximate $\mathbb{P}\big(\sum_{i=1}^{50} Y_i < 100\big)$. You may use $\Phi(0.48) \approx 0.6844$. Say which steps are exact and which are approximate.

   (c) Every Y_i is even, so $\sum Y_i$ takes only even values. Explain why $\{\sum Y_i < 100\} = \{\sum Y_i \le 98\}$ and why 99 is then the natural continuous boundary. Recompute the approximation with that boundary using $\Phi(0.38) \approx 0.6480$. The exact probability is 0.6493; which approximation is closer?

2. Let Y take the values 1, 2, 3 with probabilities 0.60, 0.10, 0.30, and let $Y_1, \ldots, Y_{100}$ be independent random variables with the same distribution as Y.

   (a) Calculate $\mathbb{E}[Y]$ and $\mathrm{Var}(Y)$.

   (b) Give the approximate distribution of $\sum_{i=1}^{100} Y_i$ and say what justifies it.

   (c) Find the approximate interquartile range of $\sum_{i=1}^{100} Y_i$. Use that the 75th percentile of the standard Normal distribution is about 0.67.

   (d) What is the approximate IQR of the sample mean $\overline{Y}_{100}$? Explain how it relates to part (c).

3. A large population has unknown mean μ and known standard deviation $\sigma = 7$. Take two independent simple random samples: one of size $n_1 = 100$ with sample mean $\overline{X}_1$, and one of size $n_2 = 400$ with sample mean $\overline{X}_2$.

   (a) Let $Y = \frac{1}{2}\overline{X}_1 + \frac{1}{2}\overline{X}_2$. Find $\mathbb{E}[Y]$ and $\mathrm{SD}(Y)$, then approximate $\mathbb{P}(Y - \mu > 0.2)$. You may use $\Phi(0.51) \approx 0.6950$.

   (b) For a number w, let $T_w = w\overline{X}_1 + (1-w)\overline{X}_2$. Show that T_w is unbiased for μ for every w.

   (c) Compare $\mathrm{Var}(T_{1/2})$ and $\mathrm{Var}(T_{1/5})$. Which estimator is preferable, and why?

4. On a hot day there are 8,000 cicadas in audible range. At any given moment a cicada either sings (probability 0.16) or does not (probability 0.84), and cicadas decide independently. The overall noise level is proportional to the number of singing cicadas.

   (a) Without computing anything, give a statistical reason to expect the fraction of singing cicadas to be about 0.16.

   (b) You listen at 9:00am. What is the approximate probability that the fraction of singing cicadas exceeds 0.17? You may use $\Phi(2.44) \approx 0.9927$.

   (c) You listen again at 9:10am, and behavior at 9:10 is independent of behavior at 9:00. What is the approximate probability that the noise at 9:10 is more than 2% larger than at 9:00? (Hint: for positive numbers A and B, $B/A > 1.02$ whenever $B - 1.02A > 0$. You may use $\Phi(0.55) \approx 0.7088$.)

5. Let $X \sim \mathrm{Binomial}(n, p)$ and define $Y = n - X$.

   (a) Is Y also binomially distributed? Explain.

   (b) Calculate $\mathrm{Cov}(X, Y)$. (The answer depends on n and p.) Explain the sign.

   (c) Let $n = 100$ and $p = 0.3$. Use a continuity-corrected Normal approximation to find $\mathbb{P}(Y - X > 30)$. (Hint: write the event in terms of X.) You may use $\Phi(0.98) \approx 0.8365$.

   (d) Repeat part (c) without a continuity correction, using 35 as the boundary (you may use $\Phi(1.09) \approx 0.8621$). The exact binomial probability is 0.8371. Compare the two approximations.

6. Let $\overline{X}_n$ be the sample mean of an independent random sample of size n from a population with mean μ and variance σ^2.

   (a) Compare the sampling distribution of $\overline{X}_n$ with the population distribution when $n = 1$ and when $n = 1000$. Discuss shape, center, and spread.

   (b) A consumer agency weighs a simple random sample of 89 flour packages. The sample mean weight is 0.902 kg and the sample standard deviation is 0.122 kg. Calculate the estimated standard error of the sample mean. How does it differ from 0.122?

(c) An employee plots a histogram of the 89 measurements, finds that it is not close to any Normal curve, and says this casts doubt on the Normal approximation for the sample mean. Comment.

7. In a population, a fraction p of individuals have $x = 0$ and a fraction $1-p$ have $x = 2$, where $0 \le p \le 1$. Let $X_1, \ldots, X_n$ be a random sample with replacement from this population.

   (a) What is the exact sampling distribution of $\overline{X}_2 = (X_1 + X_2)/2$? (The answer depends on p.)

   (b) What is the approximate sampling distribution of $\overline{X}_n$ for $n = 100$?

   (c) Is $\overline{X}_n/2$ an unbiased estimator of $1 - p$ for all $n \ge 1$? Explain.

8. Let f be a function on $[0, 1]$ with $0 \le f(x) \le 1$, and let θ be the area under f over $[0, 1]$. We want to estimate θ. Let (X, Y) be a pair of independent Uniform$(0, 1)$ random variables, so (X, Y) is a randomly chosen point in the unit square, and the probability of a region equals its area. Define

$$W = \begin{cases} 1 & \text{if } Y \le f(X), \\ 0 & \text{otherwise.} \end{cases}$$

   Let (X_i, Y_i, W_i), $i = 1, 2, \ldots$ be independent copies of (X, Y, W) and let $\overline{W}_n = \frac{1}{n}\sum_{i=1}^{n} W_i$.

   (a) What are $\mathbb{P}(W = 1)$ and $\mathbb{E}[W]$?

   (b) If n is large, is $\overline{W}_n$ a good estimator of θ? Explain.

   (c) Construct an unbiased estimator of $3\theta + 7$. What is its variance when $n = 1000$? (The answer depends on θ.)

   (d) Is $\cos(\overline{W}_n)$ an unbiased estimator of $\cos(\theta)$? Explain, and give a concrete case.

9. In a population, the quantitative variable x takes the value 0 for 20% of individuals, 2 for 15%, and 4 for 65%. A random sample with replacement of size n is drawn.

   (a) For $n = 2$, find the sampling distribution of the sample maximum M.

   (b) Is the sample maximum an unbiased estimator of the population maximum? Explain.

   (c) How large must n be so that the sampling distribution of M puts probability more than 99% on the value 4?

10. A population is described by two parameters θ_1 and θ_2. From a simple random sample of size n, an econometrician has an unbiased estimator T_1 of θ_1 and an unbiased estimator T_2 of θ_2, with variances σ_1^2 and σ_2^2.

    (a) Is $T_1 - 3T_2$ necessarily an unbiased estimator of $\theta_1 - 3\theta_2$? Explain.

    (b) Is $\mathrm{Var}(T_1 - 3T_2) = \sigma_1^2 + 9\sigma_2^2$ necessarily true? Explain.

(c) Is $T_1^2 + T_2$ necessarily an unbiased estimator of $\theta_1^2 + \theta_2$? Explain.

(d) Now suppose that for large n, T_1 and T_2 are approximately Normal and independent. Is $T_1 - 3T_2$ approximately Normal? What is its approximate distribution?

11. In a population, a fraction p of individuals have $x = 1$ and the remaining fraction $1 - p$ have $x = 3$. Let $X_1, \ldots, X_n$ be the values of x for n individuals sampled with replacement.

(a) Is $3 - \dfrac{1}{2n}\displaystyle\sum_{i=1}^{n} X_i$ an unbiased estimator of p? Explain.

(b) Is $\dfrac{9}{8} - \dfrac{1}{8n}\displaystyle\sum_{i=1}^{n} X_i^2$ an unbiased estimator of p? Explain.

(c) Suppose a single observation X is Normal with mean 1 and standard deviation σ. Construct an unbiased estimator of $7 + 4\sigma^2$ and show that it is unbiased.

12. A population consists of four individuals A, B, C, D with quantitative characteristic x equal to 0, 2, 4, 6, respectively. A simple random sample of size $n = 2$ is taken without replacement, with values X_1 and X_2.

(a) What is the sampling distribution of $Y = X_1 + X_2$?

(b) Is $Y/2$ an unbiased estimator of the population mean? Explain.

(c) Calculate $\mathrm{Var}(Y/2)$ and compare it with $\sigma^2/2$, where σ^2 is the population variance of one uniformly chosen value. What explains the difference?

13. Let X and Y be independent Normal random variables with $\mathbb{E}[X] = \mathbb{E}[Y] = \mu$, $\mathrm{SD}(X) = 2$, and $\mathrm{SD}(Y) = 5$. Consider two estimators of μ:

$$\widehat{\mu}_1 = \tfrac{1}{2}(X + Y), \qquad \widehat{\mu}_2 = 0.6X + 0.4Y.$$

(a) Which estimator is preferable? Explain.

(b) Does your comparison in part (a) depend on X and Y being Normal? Explain.